

Main

Timer 0

```

CLRF TRISC
SECF TRISC
MOVLW 0x8
MOVWF TOCON
MOVWF OXFF
MOVWF TMR0H
MOVLW 0xF2
MOVWF TMR0L
BCF INTCON, TMR0IF

```

→ BSF TOCON, TMR0ON
BSF INTCON, TMR0IE
BSF INTCON, GIE

→ *6 us delay for Main loop*
e.g. OVER MOVWF RORTC, PORTD
BRA OVER

ISR

Load values of high and lower nibble
Task kaha hai.
BCF INTCON, TMR0IF
GOTO CHR-INT

Timer 1

```

MOVLW 0x00
MOVWF T1CON
MOVLW 0xFF
MOVWF TMR1H
MOVLW 0xF2
MOVWF TMR1L
BCF PIR1, TMR1IF

```

→ BSF PIR1, TMR1IF
~~BSF PIR1,~~
BSF INTCON, PEIE
BSF T1CON, TMR1ON

→ *Jo kam Main may kaha ho us ka code likh lena.*

same as TMR0
Task kaha hai.
BCF PIR1, TMR0IF
GOTO CHR-INT

→ BTFSS INTCON, TMR0IF → Timer 0 interrupt check
Ruffic

→ BTFSS PIR1, TMR1IF → Timer 1 interrupt check.
Ruffic

→ BTFSC PIR1, TXIF → Interrupt due to Tx.

Interrupt	Flagbit	Register	Enable bit	Register
Timer 0	TMR0IF	INTCON	TMR0IE	INTCON
Timer 1	TMR1IF	PIR1	TMR1IE	PIR1
Timer 2	TMR2IF	PIR1	TMR2IE	PIR1
Timer 3	TMR3IF	PIR3	TMR3IE	PIR2

INT 0 (RB0)	INT0IF	INTCON	INT0IE	INTCON
INT 1 (RB1)	INT1IF	INTCON3	INT1IE	INTCON3
INT 2 (RB2)	INT2IF	INTCON3	INT2IE	INTCON3

PortB change interrupt
→ RB4-RB7 → single flag (RBIF)

High priority → 0008

Low priority → 0018

To make low priority of timer interrupt

```

BCF IPR1, TMR1IP
BSF IPR1, IPEN
BSF INTCON, GIEH
BSF INTCON, GIEL

```

• IPEN=0 interrupt low
directed to default location.
• IPEN=1 set low and high priority

Write in EEPROM

```

BCF EECON1, EEPGD      :: for location move
BCF EECON1, CFGS       value in EEADR
BSF EECON1, WREN       :: for data move
BCF INTCON, GIE        value in EE DATA
MOVLW 0x55
MOVWF EECON2
BSF EECON1, WR
BSF INTCON, GIE
BCF EECON1, WREN

```

Read in EEPROM

```

BCF EECON1, EEPGD      :: for location move
BCF EECON1, CFGS       value in EEADR
BSF EECON1, RD
NOP
MOVF EEDATA, PORTB

```

- ADDLW fileReg, d
- ANDLW fileReg, d
- DAW fileReg, d
- SUBLW fileReg, d
- XORLW fileReg, d
- MULLW fileReg, d
- ADDWF fileReg, d
- ADDWFC fileReg, d
- ANDWF fileReg, d
- IORWF fileReg, d
- SUBFWB fileReg, d
- SUBWF fileReg, d
- SUBWFB fileReg, d
- XORWF fileReg, d
- COMF fileReg, d
- DECF fileReg, d
- DECFSZ fileReg, d
- DECFSNZ fileReg, d
- INCF fileReg, d
- INCFNZ fileReg, d
- INCSNZ fileReg, d
- MOVF fileReg, d
- NEGF fileReg, d
- RLCF fileReg, d
- RLNCF fileReg, d
- RRCF fileReg, d
- RRNCF fileReg, d
- SWAPF fileReg, d
- BTG fileReg, d
- IORLW fileReg, d
- MOVFF fileReg, d
- BC fileReg, d
- BNC fileReg, d
- BZ fileReg, d
- BNZ fileReg, d
- BN fileReg, d
- BNC fileReg, d
- BOV fileReg, d
- BNOV fileReg, d
- BRA fileReg, d
- GOTO fileReg, d
- CALL fileReg, d
- RCALL fileReg, d
- RETURN fileReg, d
- NOP fileReg, d
- CLRF fileReg, d
- SETF fileReg, d
- BCF fileReg, bit
- BSF fileReg, bit

- BTFSS fileReg, bit
- BTFSC fileReg, bit
- BTG fileReg, bit
- STATUS
- C - Carry flag
- DC - Digital Carry flag
- Z - Zero flag
- OV - Overflow flag
- N - Negative flag
- CPFSGT
- CPFSEQ
- CPFSLT
- EQU
- ORG

- TMR0H/TMR1H
- TMR0L/TMR1L
- TMR0IF/TMR1IF
- INTCON
- PIR1

TOCON:-

- TMR0ON D7 Timer0 ON and OFF control bit
- T08BIT D6 Timer0 8-bit/16-bit selector bit
- T0CS D5 Timer0 clock source select bit
- T0SE D4 Timer0 source edge select bit
- PSA D3 Timer0 prescaler assignment bit
- TOPS2:TOPS0 D2D1D0 Timer0 prescaler selector

TICON:-

- RD16 D7 16-bit read/write enable bit
- TICKPS2:TICKPS0 D5D4 Timer1 prescaler selector
- T1OSCEN D3 Timer1 oscillator enable bit
- T1SYNC D2 Timer1 synchronization
- TMRICS D1 Timer1 clock source select bit
- TMRION DO Timer1 ON and OFF control bit